

An Abel Half-Wave Trace for Square-Billiard Clean Families

Route-A structural certificate C157

Abstract

We derive a two-dimensional Poisson formula for the genuine Dirichlet Abel half-wave trace of the unit square. Its nonaxis dual terms reorganize exactly by ordered positive primitive billiard directions and repetitions, retaining clean-family lengths and collisions. Exact shell ledgers and tail-bounded complex sentinels certify the formula. The result is a source trace and natural quantization, not an isolated-orbit determinant or target identity.

Keywords: square billiard; Abel trace; half-wave operator; Poisson summation; clean families; primitive directions.

中文摘要：本文从单位正方形的狄利克雷半波算子出发，推导其阿贝尔正则化迹的二维泊松公式。对偶格点的非轴部分可唯一整理为有序正本原台球方向与重复次数，从而完整保留光滑轨道族的长度和碰撞重数。平方范数五百以内的精确壳层账本以及带严格尾界的复数双路径计算共同验证该公式。所得对象是真实的源系统谱迹与自然量子化，但不是孤立轨道行列式，也不构成任何目标谱恒等式。

关键词：正方形台球；阿贝尔迹；半波算子；泊松求和；光滑轨道族；本原方向。

1 Dirichlet Abel trace

For the unit-square Dirichlet Laplacian, set

$$W_D(s) = \sum_{j,k \geq 1} e^{-\pi s \sqrt{j^2 + k^2}}, \quad \Re s > 0. \quad (1)$$

With the convention $\hat{f}(m) = \int_{\mathbb{R}^2} f(x) e^{-2\pi i m \cdot x} dx$, radial transformation gives

$$\widehat{e^{-\pi s |x|}}(m) = \frac{2s}{\pi(s^2 + 4|m|^2)^{3/2}}. \quad (2)$$

For real positive s , Poisson summation and $\Theta = 1 + 4/(e^{\pi s} - 1) + 4W_D$ yield

$$W_D(s) = \frac{s}{2\pi} \sum_{m \in \mathbb{Z}^2} (s^2 + 4|m|^2)^{-3/2} - \frac{1}{4} - \frac{1}{e^{\pi s} - 1}. \quad (3)$$

The primal exponential sum and dual $|m|^{-3}$ sum converge locally normally. For $\Re s > 0$, $s^2 + 4|m|^2$ avoids the nonpositive real axis, so the principal power is holomorphic; analytic continuation proves (3) on the half-plane.

2 Primitive directions and repetitions

The dual zero mode is $1/(2\pi s^2)$ and the four axes contribute

$$\frac{2s}{\pi} \sum_{r \geq 1} (s^2 + 4r^2)^{-3/2}. \quad (4)$$

Every nonaxis vector is uniquely $(\pm ra, \pm rb)$ with $a, b \geq 1$, $\gcd(a, b) = 1$. Thus the interior is

$$\frac{2s}{\pi} \sum_{\substack{a,b \geq 1 \\ (a,b)=1}} \sum_{r \geq 1} (s^2 + r^2 L_{a,b}^2)^{-3/2}, \quad L_{a,b} = 2\sqrt{a^2 + b^2}. \quad (5)$$

Four sign lifts produce the coefficient in (5), while ordered coordinate swaps remain distinct. For $s = \epsilon - it$, its principal-power branch points occur at $t = \pm r L_{a,b}$; these are the square-billiard clean-family lengths with every repetition retained.

3 Boundary strata and certified truncation

The Abel boundary contains four separately derived contributions:

1. the $m = 0$ Weyl term $1/(2\pi s^2)$;

2. axis $-3/2$ branches at $t = \pm 2r$;
3. interior clean-family $-3/2$ branches at $t = \pm rL_{a,b}$;
4. simple poles of $-1/(e^{\pi s} - 1)$ at $s = 2iq$, equivalently $t \in 2\mathbb{Z}$.

Axis branches may coincide with boundary-subtraction poles, but their singularity types differ and no cancellation is asserted. In particular, the branch list is not presented as the complete boundary singular set.

For numerical verification, if the primal box is $1 \leq j, k \leq J$, then

$$|R_J| \leq \frac{2q^{J+2}}{(1-q)^2}, \quad q = e^{-\pi \Re(s)/\sqrt{2}}. \quad (6)$$

For an integer max-norm cutoff $M \geq |s|$, subtract on the dual side $1/(8|m|^3) - 3s^2/(64|m|^5)$ and add the exact Epstein sums $4\zeta(\alpha)\beta(\alpha)$ for $\alpha = 3/2, 5/2$. Taylor's theorem and the $8k$ points of max norm k bound the accelerated tail by

$$\frac{|s|^5}{2\pi 3^{5/2} M^5}. \quad (7)$$

4 Certificate and Route-A boundary

Through squared norm 500, the exact ledger contains 98 primitive shells, 239 ordered primitive directions, 161 occupied dual shells, and 373 ordered positive vectors. The first fourfold primitive collision is

$$(1, 8), (4, 7), (7, 4), (8, 1), \quad a^2 + b^2 = 65.$$

At $s = 0.9 + 0.4i$ and $s = 1.3 + 0.7i$, two complex sentinels compare (1) against an Epstein-accelerated form of (3), with respective absolute differences 5.18×10^{-13} and 3.92×10^{-12} , reported beside explicit analytic truncation bounds. The deterministic 55-digit centers are not interval-arithmetic outputs.

The tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION). Although $W_D = \text{Tr } e^{-s\sqrt{\Delta_D}}$ is a genuine source trace, the periodic sets are clean families. We claim no isolated-orbit determinant or stability amplitude, target trace/divisor/functional equation or counting law, arithmetic local/Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

Data availability. All claim-bearing data are the exact shell JSON and scripts in this C157 package; no external dataset is used. **Ethics.** The work uses no human participants, animals, or sensitive personal data. **Author contributions/CReditT.** This is an anonymous technical certificate with no individual authorship assignment; package provenance records Conceptualization, Formal analysis, Software, Validation, and Writing roles. **Competing interests.** None known. **Funding.** No external funding is reported. **AI-use disclosure.** An AI coding assistant supported drafting, derivation checks, and code review. Exact scripts and independent paths reproduce every quantitative claim; the assistant was not treated as an external peer reviewer.